

Title: LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval

Abstract

Customer support systems play a critical role in improving user satisfaction by providing timely and accurate responses to customer queries. Traditional chatbot systems rely on predefined rules or keyword matching, making them ineffective in handling complex, context-aware conversations and retrieving relevant information from large knowledge bases. This project proposes an **LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval**, which combines Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) techniques to deliver accurate, context-aware, and explainable responses to customer inquiries.

The proposed system uses state-of-the-art language models such as **Llama 3, Mistral, or Google's Gemini**, integrated with a vector database to retrieve relevant information from organizational documents, FAQs, product manuals, and customer support knowledge bases. Customer queries are first converted into embeddings, and the most relevant documents are retrieved using semantic similarity search. The retrieved context is then supplied to the LLM, enabling it to generate accurate and grounded responses while minimizing hallucinations.

The chatbot supports multi-turn conversations by maintaining conversation history and contextual understanding. It also incorporates confidence estimation and source attribution, allowing users to view the documents or knowledge sources used to generate each response. Additional features such as conversation summarization, intent detection, sentiment analysis, and escalation of unresolved queries to human support agents can further enhance customer service efficiency.

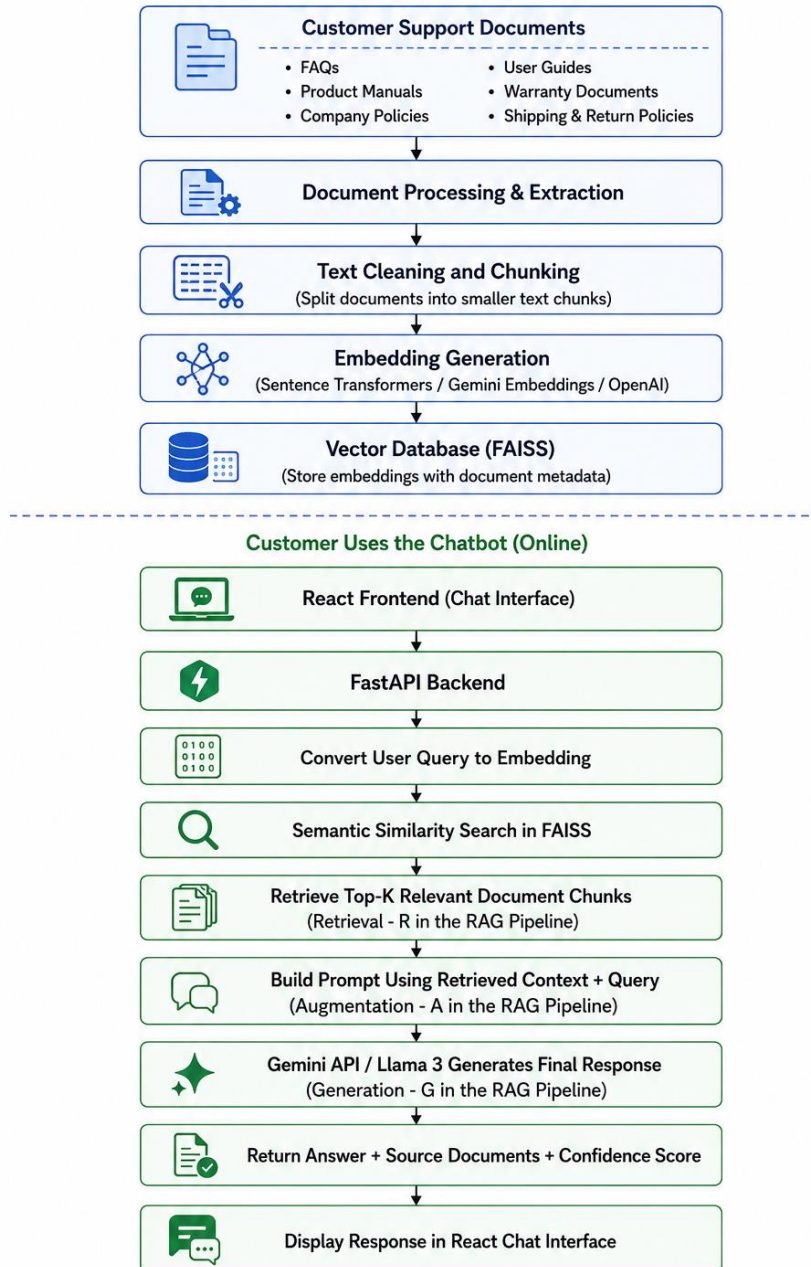
The performance of the proposed system is evaluated using retrieval accuracy, response relevance, answer correctness, BLEU score, ROUGE score, semantic similarity metrics, response latency, and user satisfaction. The resulting framework provides an intelligent, scalable, and reliable customer support solution that improves customer experience while reducing manual workload and operational costs.

Datasets

Dataset	URL link
<u>Customer Support FAQ Dataset</u>	https://huggingface.co/datasets
<u>Amazon Customer Reviews</u>	https://huggingface.co/datasets/mteb/amazon-reviews_mult
<u>Customer Support on Twitter</u>	https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter

Approach

LLM-Based Intelligent Customer Support Chatbot using Retrieval-Augmented Generation (RAG)



Project Group Members

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